Variables

```
name = "Sathya"
print(name)
```

```
Sathya
```

```
a = 10
b = 20
a, b = b, a
print(a, b)
```

```
20 10
```

```
x, y, z = 1, 2, 3
print(x + y + z)
```

```
6
```

```
num = 100
print("Before:", num)
num = 200
print("After:", num)
```

```
Before: 100
After: 200
```

```
val = 50
del val
print(val)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_9219/1220538186.py in <cell line: 0>()
      1 val = 50
      2 del val
----> 3 print(val)

NameError: name 'val' is not defined
```

Next steps: Explain error

Data Types

```
a = 10
b = 3.14
```

```
c = "Hello"
d = True

print(type(a), type(b), type(c), type(d))
```

```
<class 'int'> <class 'float'> <class 'str'> <class 'bool'>
```

```
print(type(3.14), type("Hello"), type(True), type(0))
```

```
<class 'float'> <class 'str'> <class 'bool'> <class 'int'>
```

```
x = 9
print(isinstance(x, int))
```

```
True
```

```
print(type(None))
```

```
<class 'NoneType'>
```

```
big_int = 12345678901234567890
small_float = 0.0000001
print(big_int, small_float)
```

```
12345678901234567890 1e-07
```

Type Conversion

```
num = int("123")
print(num + 7)
```

```
130
```

```
print(int(9.99))
```

```
9
```

```
print(bool(0), bool(1))
```

```
False True
```

```
print(int(True), int(False))
```

```
1 0
```

```
user_input = float(input("Enter number: "))
print(user_input * 2)
```

```
Enter number: 2
4.0
```

```
int("hello")
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_9219/3952165420.py in <cell line: 0>()
----> 1 int("hello")

ValueError: invalid literal for int() with base 10: 'hello'
```

Next steps: Explain error

String Manipulation

```
text = "Hello, World!"
print(text.upper())
print(text.lower())
```

```
HELLO, WORLD!
hello, world!
```

```
print(len("Python Programming"))
```

```
18
```

```
text = "Hello, World!"
print(text[7:12])
```

```
World
```

```
print("abcdef"[::-1])
```

```
fedcba
```

```
print("nan" in "banana")
```

```
True
```

```
print("This is a bad day".replace("bad", "good"))
```

```
This is a good day
```

```
print(" hello ".strip())
```

```
hello
```

```
print("apple,banana,cherry".split(","))
```

```
['apple', 'banana', 'cherry']
```

```
print("banana".count("a"))
```

```
3
```

```
print("Python123".isalnum())
```

```
True
```

String Formatting

```
name = "John"
age = 25
print(f"My name is {name} and I am {age} years old")
```

```
My name is John and I am 25 years old
```

```
print("The price is {} dollars".format(49.99))
```

```
The price is 49.99 dollars
```

```
print(f"{3.14159:.2f}")
```

```
3.14
```

```
print(f"{7:03}")
```

```
007
```

```
a = 5
b = 3
print(f"{a} x {b} = {a*b}")
```

```
5 x 3 = 15
```

```
names = ["Alice", "Bob"]
scores = [85, 90]

for i in range(len(names)):
    print(f"{names[i]:<10} {scores[i]:>5}")
```

```
Alice         85
Bob           90
```

Python Lists

```
fruits = ["apple", "banana", "cherry", "mango", "orange"]
print(fruits)
```

```
['apple', 'banana', 'cherry', 'mango', 'orange']
```

```
nums = [10, 20, 30, 40, 50]
print(nums[0], nums[-1])
```

```
10 50
```

```
fruits.append("grape")
print(fruits)
```

```
['apple', 'banana', 'cherry', 'mango', 'orange', 'grape']
```

```
nums = [1, 2, 3, 4]
nums.insert(2, 99)
print(nums)
```

```
[1, 2, 99, 3, 4]
```

```
fruits = ["apple", "banana", "cherry"]
fruits.remove("banana")
print(fruits)
```

```
['apple', 'cherry']
```

```
nums = [5, 2, 8, 1, 9]
print(sorted(nums))
print(sorted(nums, reverse=True))
```

```
[1, 2, 5, 8, 9]
[9, 8, 5, 2, 1]
```

```
nums = [3, 7, 1, 9, 4]
print(len(nums), max(nums), min(nums))
```

```
5 9 1
```

```
nums = [0, 1, 2, 3, 4, 5, 6]
print(nums[2:5])
```

```
[2, 3, 4]
```

```
original = [1, 2, 3]
copy = original.copy()
copy.append(4)
print(original, copy)
```

```
[1, 2, 3] [1, 2, 3, 4]
```

```
list1 = [1, 2, 3]
```

```
list1 = [1, 2, 3]
list2 = [4, 5, 6]
merged = list1 + list2
print(merged)
```

```
[1, 2, 3, 4, 5, 6]
```